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Q1. Smart Hospital Patient Registration System

Score: 0.00 **TC: 0/3 passed**

CODE

```
import java.util.*;

public class Main {
    static ArrayList<String> patientIds = new ArrayList<>();
    static HashMap<String, String[]> patientMap = new HashMap<>();
    static HashMap<String, Integer> deptCount = new HashMap<>();
    static final int MAX_PATIENTS = 100;

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        boolean running = true;

        while (running) {
            System.out.println("1. Register Patient");
            System.out.println("2. Search Patient");
            System.out.println("3. Display All Patients");
            System.out.println("4. Count Department-wise Patients");
            System.out.println("5. Exit");

            int choice = Integer.parseInt(sc.nextLine().trim());

            switch (choice) {
                case 1:
                    registerPatient(sc);
                    break;
                case 2:
                    searchPatient(sc);
                    break;
                case 3:
                    displayAllPatients();
                    break;
                case 4:
                    departmentWiseCount();
                    break;
                case 5:
                    running = false;
                    break;
                default:
                    System.out.println("Invalid Choice");
            }
        }
        sc.close();
    }

    static void registerPatient(Scanner sc) {
        if (patientIds.size() >= MAX_PATIENTS) {
            System.out.println("Maximum Patient Limit Reached.");
            return;
        }

        String id = sc.nextLine().trim();

        if (patientMap.containsKey(id)) {
            System.out.println("Duplicate Patient ID. Registration Failed.");
            return;
        }

        String name = sc.nextLine().trim();
        if (!name.matches("[a-zA-Z ]+")) {
            System.out.println("Invalid Name. Only alphabets allowed.");
            return;
```

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        }

        String mobile = sc.nextLine().trim();
        if (!mobile.matches("\\d{10}")) {
            System.out.println("Invalid Mobile Number. Must be 10 digits.");
            return;
        }

        String department = sc.nextLine().trim().toUpperCase();
        String doctor = sc.nextLine().trim();

        String properName = toProperCase(name);

        String[] details = {id, properName, mobile, department, doctor};
        patientIds.add(id);
        patientMap.put(id, details);

        deptCount.put(department, deptCount.getOrDefault(department, 0) + 1);

        System.out.println("Patient Registered Successfully.");
    }

    static void searchPatient(Scanner sc) {
        String id = sc.nextLine().trim();

        if (patientMap.containsKey(id)) {
            String[] p = patientMap.get(id);
            System.out.println("Patient Details");
            System.out.println("ID : " + p[0]);
            System.out.println("Name : " + p[1]);
            System.out.println("Mobile : " + p[2]);
            System.out.println("Department : " + p[3]);
            System.out.println("Doctor : " + p[4]);
        } else {
            System.out.println("Patient Not Found");
        }
    }

    static void displayAllPatients() {
        System.out.println("----------------------------------------------------");
        System.out.println("ID\tName\tMobile\tDepartment\tDoctor");
        System.out.println("----------------------------------------------------");
        for (String id : patientIds) {
            String[] p = patientMap.get(id);
            System.out.println(p[0] + "\t" + p[1] + "\t" + p[2] + "\t" + p[3] + "\t" + p[4]);
        }
    }

    static void departmentWiseCount() {
        System.out.println("Department Count");
        for (Map.Entry<String, Integer> entry : deptCount.entrySet()) {
            System.out.println(entry.getKey() + " : " + entry.getValue());
        }
    }

    static String toProperCase(String name) {
        String[] words = name.trim().split("\\s+");
        StringBuilder sb = new StringBuilder();
        for (String word : words) {
            if (word.length() > 0) {
                sb.append(Character.toUpperCase(word.charAt(0)))
                  .append(word.substring(1).toLowerCase())
                  .append(" ");
            }
        }
```

```
        return sb.toString().trim();
    }
}
```

OUTPUT